

P241: Foundations of Artificial Intelligence Ethics

Indiana University, Bloomington, Spring 2027

Course Website: calum-mcnamara.github.io/ai-ethics

Administrative Information

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Lecture Times: TBD

Lecture Location: TBD

Course Description

Unless you've been living under a rock, you'll be aware of the profound impact that AI technologies are having on society. Increasingly, AI models are being given the authority to make decisions of great importance, without direct human oversight. This, in turn, raises various ethical issues. In this course, we will discuss such ethical issues, focusing in particular on two connected problems raised by modern AI: interpretability and alignment. In the first part of the course, we will ask what it means to "understand" an AI system, paying special attention to answers provided by the emerging science of *mechanistic interpretability*. In the second part of the course, we will then investigate what it means for an AI system to be "aligned" with human interests and values. Throughout, our discussions will be underpinned by classic theories from ethics, the philosophy of mind, the philosophy of science, and political philosophy. In the final weeks of the semester, students will have the opportunity to collaborate with one another on an open-ended research project.

For this course, no prior background in AI or philosophy (or math, or computer science, etc.) will be assumed. However, students will be expected to work hard to acquire bits of this background along the way.

Assessment and Grading

- Attendance and Participation: 10%
- Midterm Exam: 30%
- Final Project Prospectus: 10%
- Final Project Submission: 40%
- Final Project Oral Defense: 10%

The Final Project

This course will culminate in a collaborative research project, which students will complete in small groups (3-5 people). The project is deliberately open-ended and may take a variety of forms. For example, a group might replicate or extend the results of a research paper, conduct an original empirical or interpretability study, develop a technical prototype, formulate and defend a policy proposal, or write a philosophical research article, investigating an ethical question raised by AI. The aim of this project is to give students an opportunity to pursue an important question in depth, using the methods most appropriate to it. Projects should be intellectually ambitious but feasible within the time available, and groups will work with the instructional team to determine an appropriate format and scope. At the end of the course, an outstanding project—as determined by the instructional team—will receive a prize. Especially promising projects may also be developed further for submission to an appropriate conference or publication venue. (The bar for this will be high.)

Students should form project-groups by the end of Week 9. At the end of Week 11, each group will submit a prospectus outlining its topic, central question, and proposed approach (worth 10% of the final grade). After that, the final three weeks of the course (weeks 13–15) will function as an in-class project studio. Groups will use this time to refine their research question, divide responsibilities, consult with the instructional team, and present work in progress to the class for feedback. The project will culminate in a final submission and an oral defense, during which time students will explain and defend their central claims, and reflect on the choices they made as a group. The very last day of classes will be a poster presentation session, in which students will have the opportunity to present their work to one another.

Discussion Sections

Reasoning responsibly about AI ethics requires us to be familiar with how AI systems actually work. For this reason, discussion sections in this class are going to be a bit unusual. Specifically, rather than following the “standard” format for philosophical discussion sections, much of our time will be spent working through simple coding notebooks with the instructors. These exercises will give students the opportunity to tinker with AI models and to apply various interpretability

and alignment techniques, rather than merely reading about them in the abstract. No prior programming experience will be assumed, and students will be graded only on their attendance and participation in the sections, and not on their coding ability. The notebooks will introduce all the necessary concepts and code incrementally, and students will work through them collaboratively, with guidance from the instructional team. Discussion sections will also provide time to connect the results of these exercises to the philosophical questions raised in lectures and the assigned readings.

Readings and Lecture Content

The assigned readings for this course are intended to supplement, rather than duplicate, the material covered in lectures. So, although the readings and lectures will address many of the same topics, the overlap between them will not be exact: lectures may introduce arguments, examples, or technical details not contained in the readings (and will frequently be broader than the assigned readings), while the readings may develop issues that receive less attention in class. Students are responsible for learning material presented in both formats.

AI Use and Academic Integrity

Students are *allowed* to use generative AI tools in this class, to *aid* them in completing the collaborative final project. Note, however, that the key word here is ‘aid’. AI tools may be used, for example, to brainstorm ideas, clarify technical concepts, assist with coding or debugging, formulate possible objections, or improve the organization and presentation of the project. But they may *not* substitute for the group’s own research, reasoning, or writing. The project’s central intellectual decisions and substantive contributions *must* be made by members of the group. (For what it’s worth, I don’t think AI models are typically very good at the last things here anyway.)

Students remain responsible for the accuracy, originality, and integrity of everything they submit, including any claims, citations, code, or data produced with AI assistance. Every member of the group must be able to—and indeed, will ultimately have to—explain and defend every part of their project during the oral defense. All uses of AI *must* be disclosed in an AI Use Statement submitted with the project. This statement should identify the tools that were used and briefly explain how they were used.

Undisclosed or otherwise unauthorized use of AI may constitute academic misconduct. If you are uncertain whether a particular use is acceptable, please ask the instructional team before submitting your work.

For definitions and examples of academic misconduct, see the [IU Code of Student Rights, Responsibilities, and Conduct](#). For information about Bloomington procedures and potential sanctions, see [Academic Misconduct Procedures](#).

Technology in Lectures

With two exceptions, laptops, phones, and other electronic devices are not permitted to be used during lectures. My reason for this is that there’s good evidence to suggest that laptops in lecture

lower exam performance and overall engagement. For example, in a large randomized study at West Point, it was found that allowing laptops and tablets in class reduced students' final exam scores ([Carter et al. 2017](#)). And other research has found that laptop multitasking distracts not only the user, but also students sitting nearby ([Sana et al. 2013](#)).

The first exception to this rule is if you're going to take notes *by hand* on a device like an iPad or a reMarkable. That's allowed. The other exception is if you have a disability that necessitates using a computer, or some other electronic device. Obviously, if that's the case, then I'm happy to accommodate you. But if this is so, please speak to me about it as early as you can.

Communication

If you have a question about course mechanics or housekeeping, please first check the syllabus. If, after having checked the syllabus, your question still isn't answered, please try to stop by office hours (either mine—Prof McNamara's—or your TA's). If you can't do this, or if your question is urgent, then you should feel free to email us. If you do so, then we will endeavor to reply within 24 hours. The exception to this rule is if you email on a Friday. In that case, we will try to answer by the end of the day the following Monday. Please do *not* repeat-email us unless you haven't heard back from us within these time frames.

Disabilities

The Americans with Disabilities Act is a federal anti-discrimination statute that provides comprehensive civil rights protection for persons with disabilities. Students who require academic accommodations can work with AES (Accessible Educational Services) to arrange for (among other things) assistive technology or academic coaching. Many students don't receive adequate diagnoses, or discover only late in their academic careers that they have access to academic accommodations. Thus, if you even *suspect* that you might need such accommodations, it's worth reaching out early on.

For reasons of privacy and consistency, I ask that you start with AES; in particular, see [here](#) for further details. However, please know that whether or not you are able to arrange formal accommodations, I am committed to working with you to ensure that you are able to participate fully in this course. Just speak to me about your needs, as soon as you can.

Mental Health and “Ghosting”

In the past, I've had students who stop coming to class or section, and stop handing in assignments, partway through the term. This is usually because the student is overwhelmed with work, facing a difficult personal situation, or struggling with a mental health issue. If you find yourself confronted with any of these issues (or a similar one), then I strongly encourage you to talk to the university's counseling service as soon as you can: see [here](#) for contact information. They are professionals and they care very much about your well-being. I also encourage you to communicate with me or your TA about any missing work. We're not mental health professionals, but we care about your

well-being, and can better help you manage your assignments if we know what's going on as it's happening, instead of being informed of any issues shortly before (or after) the end of term.

Tentative Schedule

Part I: Understanding Artificial Intelligence

Week 1: Introduction

Day One: Course Overview

Optional Readings:

- Amodeli, [“The Urgency of Interpretability”](#) (2025)
- OpenAI, [“Hugging Face Incident and the Road Ahead”](#) (2026)
- METR, [“OpenAI-Hugging Face Incident Investigation”](#) (2026)

Day Two: A Brief History of AI, and the Path Forward

Primary Readings:

- Turing, [“Computing Machinery and Intelligence”](#) (1950), Sections 1, 6, and 7
- Bostrom, *Superintelligence* (2014), Chapter 1

Optional Readings:

- Good, [“Speculations Concerning the First Ultraintelligent Machine”](#) (1965)

Week 2: How Do AI Models Work?

Day One: Neural Networks and Transformers

Primary Readings:

- Levinstein, [“A Conceptual Guide to Transformers: Part I”](#) (2024)
- Levinstein, [“A Conceptual Guide to Transformers: Part II”](#) (2024)

Optional Readings:

- Nielsen, *Neural Networks and Deep Learning* (2019), Chapter 1 (encouraged)
- 3Blue1Brown, [Essence of Linear Algebra](#) (video series)

Day Two: Training and Learning

Primary Readings:

- Levinstein, [“A Conceptual Guide to Transformers: Part III”](#) (2024)
- Levinstein, [“A Conceptual Guide to Transformers: Part IV”](#) (2024)

Optional Readings:

- Nielsen, *Neural Networks and Deep Learning* (2019), Chapter 2

- 3Blue1Brown, [Neural Networks](#) (video series)

Week 3: The Science of Interpretability

Day One: Interpretability: Aims and Methods

Primary Readings:

- Lipton, [“The Mythos of Model Interpretability”](#) (2018), Sections 1–3
- Herrmann and Levinstein, [Radical AI Interpretability](#) (2026), Preface (“A Parable”) and Chapter 1

Optional Readings:

- Rudin, [“Stop Explaining Black Box Machine Learning Models for High Stakes Decisions and Use Interpretable Models Instead”](#) (2019), pp. 206–210

Day Two: Evaluating Interpretability Methods

Primary Readings:

- Doshi-Velez and Kim, [“Towards a Rigorous Science of Interpretable Machine Learning”](#) (2017), Sections 1–4
- Adebayo et al., [“Sanity Checks for Saliency Maps”](#) (2018), Sections 1–3

Optional Readings:

- Jacovi and Goldberg, [“Towards Faithfully Interpretable NLP Systems: How Should We Define and Evaluate Faithfulness?”](#) (2020)

Week 4: Mechanistic Interpretability: Part I

Day One: Circuits and Representations

Primary Readings:

- Olah et al., [“Zoom In: An Introduction to Circuits”](#) (2020)
- Herrmann and Levinstein, [Radical AI Interpretability](#) (2026), Chapter 3

Optional Readings:

- Elhage et al., [“A Mathematical Framework for Transformer Circuits”](#) (2021), “Transformer Overview” section

Day Two: Causal Methods and Circuit Discovery

Primary Readings:

- Wang et al., [“Interpretability in the Wild: A Circuit for Indirect Object Identification in GPT-2 Small”](#) (2023), Sections 1–4
- Geiger et al., [“Causal Abstraction: A Theoretical Foundation for Mechanistic Interpretability”](#) (2025), Sections 1–2

Optional Readings:

- Conmy et al., [“Towards Automated Circuit Discovery for Mechanistic Interpretability”](#) (2023), Sections 1–2

Week 5: Mechanistic Interpretability: Part II

Day One: Features, Superposition, and Sparse Autoencoders

Primary Readings:

- Elhage et al., [“Toy Models of Superposition”](#) (2022), “Overview,” “Definitions and Motivation,” and “Toy Models”
- Bricken et al., [“Towards Monosemanticity: Decomposing Language Models with Dictionary Learning”](#) (2023), “Introduction,” “Problem Setup,” and “Results”

Optional Readings:

- Olah, [“Mechanistic Interpretability, Variables, and the Importance of Interpretable Bases”](#) (2022)

Day Two: Applications and the Limits of Mechanistic Interpretability

Primary Readings:

- Chalmers, [“Propositional Interpretability in Artificial Intelligence”](#) (2025), Sections 1, 5, and 7
- Herrmann and Levinstein, [Radical AI Interpretability](#) (2026), Sections 6.1–6.3

Optional Readings:

- Sharkey et al., [“Open Problems in Mechanistic Interpretability”](#) (2025), Sections 1–2

Week 6: Review and Midterm

Day One: Review Session

Day Two: Midterm Exam

Part II: Aligning Artificial Intelligence

Week 7: The Alignment Problem

Day One: What Is Alignment?

Primary Readings:

- Wiener, [“Some Moral and Technical Consequences of Automation”](#) (1960)
- Gabriel, [“Artificial Intelligence, Values, and Alignment”](#) (2020), Sections 1–3

Optional Readings:

- Hadfield-Menell and Hadfield, [“Incomplete Contracting and AI Alignment”](#) (2018), Sections 1–3

Day Two: Why Is Alignment Difficult?

Primary Readings:

- Ngo, Chan, and Mindermann, [“The Alignment Problem from a Deep Learning Perspective”](#) (2022), Sections 1–2

- Shah et al., [“Goal Misgeneralization: Why Correct Specifications Aren’t Enough for Correct Goals”](#) (2022), Sections 1–3

Optional Readings:

- Amodei et al., [“Concrete Problems in AI Safety”](#) (2016), Sections 1–3

Week 8: Values and Preferences

Day One: Preferences, Welfare, and Human Interests

Primary Readings:

- Parfit, *Reasons and Persons* (1984), Appendix: “What Makes Someone’s Life Go Best?”

Optional Readings:

- Zhi-Xuan et al., [“Beyond Preferences in AI Alignment”](#) (2024), Sections 1–3
- Sen, [“Rational Fools: A Critique of the Behavioural Foundations of Economic Theory”](#) (1977), Sections I–III

Day Two: Value Pluralism and Moral Uncertainty

Primary Readings:

- Berlin, [“The Pursuit of the Ideal”](#) (1988)
- MacAskill, Bykvist, and Ord, [Moral Uncertainty](#) (2020), Introduction and Chapter 1

Optional Readings:

- Sen, [“The Impossibility of a Paretian Liberal”](#) (1970)

Week 9: RLHF and Constitutional AI

Day One: Reinforcement Learning from Human Feedback

Primary Readings:

- Ouyang et al., [“Training Language Models to Follow Instructions with Human Feedback”](#) (2022), Sections 1–3
- Casper et al., [“Open Problems and Fundamental Limitations of Reinforcement Learning from Human Feedback”](#) (2023), Sections 1–2

Day Two: Constitutional AI

Primary Readings:

- Bai et al., [“Constitutional AI: Harmlessness from AI Feedback”](#) (2022), Sections 1–3
- Anthropic, [“Claude’s Constitution”](#) (2026)

Optional Readings:

- Huang et al., [“Collective Constitutional AI: Aligning a Language Model with Public Input”](#) (2024)

Week 10: Emergent Misalignment and Evaluations

Day One: Emergent Misalignment

Primary Readings:

- Betley et al., [“Training Large Language Models on Narrow Tasks Can Lead to Broad Misalignment”](#) (2026)

Optional Readings:

- Wang et al., [“Persona Features Control Emergent Misalignment”](#) (2025)
- Del Pinal et al., [“Emergent Alignment and the Projectability of Ethical Personas”](#) (manuscript)

Day Two: Guest Lecture

Optional Readings:

- Shevlane et al., [“Model Evaluation for Extreme Risks”](#) (2023), Sections 1–3

Week 11: Control and Scalable Oversight

Day One: AI Control and Corrigibility

Primary Readings:

- Greenblatt and Shlegeris, [“The Case for Ensuring That Powerful AIs Are Controlled”](#) (2024)
- Korbak et al., [“How to Evaluate Control Measures for LLM Agents? A Trajectory from Today to Superintelligence”](#) (2025), Sections 1–3
- Hadfield-Menell et al., [“The Off-Switch Game”](#) (2017)

Optional Readings:

- Greenblatt et al., [“AI Control: Improving Safety Despite Intentional Subversion”](#) (2023), Sections 1–3

Day Two: The Supervision Gap and Scalable Oversight

Primary Readings:

- Bowman et al., [“Measuring Progress on Scalable Oversight for Large Language Models”](#) (2022), Sections 1–3

Optional Readings:

- Burns et al., [“Weak-to-Strong Generalization: Eliciting Strong Capabilities with Weak Supervision”](#) (2023), Sections 1–3

Week 12: Governing Artificial Intelligence

Day One: The Limits of Technical Alignment

Primary Readings:

- Gabriel and Keeling, [“A Matter of Principle? AI Alignment as the Fair Treatment of Claims”](#) (2025), Sections 1–3

Optional Readings:

- Mittelstadt, [“Principles Alone Cannot Guarantee Ethical AI”](#) (2019)

Day Two: Policy, Governance, and International Coordination

Primary Readings:

- AI Futures Project, [“AI 2040: Summary of Plan A”](#) (2026)
- Anderljung et al., [“Frontier AI Regulation: Managing Emerging Risks to Public Safety”](#) (2023), Sections 1–3

Optional Readings:

- Erman and Furendal, [“Artificial Intelligence and the Political Legitimacy of Global Governance”](#) (2024)

Part III: Collaborative Research Projects

Week 13: Project Studio I

Day One: Project Workshop

Day Two: Project Workshop

Week 14: Project Studio II

Day One: Project Workshop

Day Two: Final Project Submission Due

Week 15: Project Studio III

Day One: Oral Defenses

Day Two: Poster Presentations